How-To Library

Informatica

Using Nagios Core for Monitoring in Dynamic Data Masking

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Abstract

You can use Nagios Core to monitor Informatica® Dynamic Data Masking servers on Windows and Linux machines. This article provides instructions for installing and configuring Nagios Core for monitoring Dynamic Data Masking.

Supported Versions

- Dynamic Data Masking 9.9.2

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Overview

Nagios Core is an open source system and network monitoring tool. It monitors hosts and services, generating email alerts when issues arise and when issues are rectified.

You can configure Nagios Core to monitor a Dynamic Data Masking server on Linux or Windows.

On Linux, enter the directories where Dynamic Data Masking is installed in the configuration file. Install the custom plugin. The plugin notifies the Nagios Core server of the corresponding directories if any of the services go down.

On Windows, configure the required Dynamic Data Masking services. NSClient++ monitors the services and notifies the Nagios Core server if a service goes down.

Perform the following steps to install and configure Nagios Core for monitoring Dynamic Data Masking on a Linux machine:

1. Install Nagios Core server on a Linux machine.
2. Install the Nagios Remote Plugin Executor on Linux host machines.
3. Configure the services required for monitoring in Linux host machines. This can include the Nagios Core machine.
4. Configure the Dynamic Data Masking server for monitoring.

Perform the following steps to install and configure Nagios Core for monitoring Dynamic Data Masking on a Windows machine:

1. Install Nagios Core server on a Linux machine.
2. Install NSClient++ on each Windows machine with Dynamic Data Masking.
3. Configure the Dynamic Data Masking server for monitoring.

You can set up email notification services in the Nagios Core server to generate and send notifications.

Configuring Nagios Core for Dynamic Data Masking on Linux

You can configure Nagios Core to monitor Dynamic Data Masking that you install on Linux machines.

The following sections describe the steps to configure Nagios Core to monitor Dynamic Data Masking on Linux machines. A single Nagios Core server can monitor multiple Dynamic Data Masking services.

Step 1. Install Nagios Core Server on a Linux Machine

To use Nagios Core for monitoring, you must install the Nagios Core server on a Linux machine.

You must have installed CentOS, Red Hat Enterprise Linux, or Fedora systems before you begin the steps. Log in as a root user to perform the install steps.

1. Run the following commands to install Apache and PHP and start the Apache service:

   ```
   #yum install httpd php php-cli gcc unzip wget glibc glibc-common gd gd-devel net-snmp
   #yum groupinstall "development tools"
   #service httpd start
   ```

2. Run the following commands to create a Nagios Core user account and set the password:

   ```
   #useradd nagios
   #passwd nagios
   #groupadd nagcmd
   #usermod -a -G nagcmd nagios
   #usermod -a -G nagcmd apache
   ```

   The code contains a sample user ID and password. Replace your User ID and password in the commands.

3. Run the following commands to download and install the latest Nagios Core service from the official web site:

   ```
   # cd /opt/
   # wget https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.4.5.tar.gz
   # tar xzf nagios-4.4.5.tar.gz
   # cd nagios-4.4.5
   # ./configure --with-command-group=nagcmd
   # make all
   # make install
   # make install-init
   # make install-daemoninit
   # make install-config
   # make install-commandmode
   # make install-exfoliation
   ```

4. Run the following command to configure Apache for Nagios Core:

   ```
   #make install-webconf
   #htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
   #service httpd restart
   ```

5. Run the following commands to install Nagios plugins:

   ```
   #cd /opt
   #wget http://nagios-plugins.org/download/nagios-plugins-2.2.1.tar.gz
   #tar xzf nagios-plugins-2.2.1.tar.gz
   #cd nagios-plugins-2.2.1
   ```

```
#./configure --with-nagios-user=nagios --with-nagios-group=nagios
#make
#make install
#/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
#service nagios start
```

Step 2. Install Nagios Remote Plugin Executor

The Nagios Remote Plugin Executor (NRPE)allows you to remotely run Nagios plugins on Linux machines.

Log in as a root user to perform the installation. NRPE uses port 5666. The port must be open on the local firewall.

1. Run the following commands to download the source compile and create a user and group. The command also installs the binaries and configuration files.

```
#yum install -y gcc glibc glibc-common openssl openssl-devel perl wget
#cd /root/nagios
#wget --no-check-certificate -O nrpe.tar.gz https://github.com/NagiosEnterprises/nrpe/
archive/nrpe-3.2.1.tar.gz
#tar xzf nrpe.tar.gz
#cd /root/nagios/nrpe-nrpe-3.2.1/
#./configure --enable-command-args
#make all
#make install-groups-users
#make install
#make install-config
#echo >> /etc/services
#echo '# Nagios services' >> /etc/services
#echo 'nrpe    5666/tcp' >> /etc/services
#make install-init
#systemctl enable nrpe.service
#firewall-cmd --zone=public --add-port=5666/tcp
#firewall-cmd --zone=public --add-port=5666/tcp --permanent
```

2. Open the following file: `/usr/local/nagios/etc/nrpe.cfg`
   a. To enable Nagios Core to connect to another host, add the IP address separated by a comma.

      For example: `allowed_hosts=127.0.0.1,10.25.5.2`
      In the example, the first IP address is of the local host and the second address is the IP address of the Nagios Core server.

   b. Configure the following property to enable advanced NRPE configuration:

      `dont_blame_nrpe=1`
      The option determines whether the NRPE daemon allows clients to specify arguments to commands that are executed.

3. Run the following command to start the NRPE service:

```
#systemctl start nrpe.service
```

Step 3. Configure Services for Monitoring

After you configure the Nagios Remote Plugin Executor, add the required services on the Nagios Core machine.

1. Browse to the following directory and verify that the `check_nrpe` file is present.

```
cd /usr/local/nagios/libexec/
```

2. If the file is not present, copy the file from the following Nagios path:

```
cp /opt/nagios-4.4.5/nrpe-3.2.1/src/check_nrpe /usr/local/nagios/libexec/
```

3. To verify that Nagios Core can connect to the Linux client machine, run the following command from the `/usr/local/nagios/libexec/`directory:

```
./check_nrpe -H <IP address of the client machine where nrpe is installed>
```

```
[root@<host> objects]# cd /usr/local/nagios/libexec/
[root@<host> libexec]# ./check_nrpe -H <ip address>
NRPE v3.2.1
[root@<host> libexec]#
```

If you do not get a result, check if you can successfully ping the client machine, and that ports 5666, 5667, and 5668 are open on the local firewall of the client machine.

4. Create a configuration file for Nagios core to use to monitor the Linux client machine. Browse to the following directory:

   ```
   cd /usr/local/nagios/etc/objects
   ```

5. Create a copy of the `localhost.cfg` Nagios sample file for Linux.

   ```
   cp localhost.cfg remoteClient.cfg
   ```

6. Add an entry for the `remoteClient.cfg` file to the main Nagios configuration file. Open the `nagios.cfg` file.

   ```
   vi /usr/local/nagios/etc/nagios.cfg
   ```

7. Add the files created:

   ```
   #Definitions for monitoring the local (Linux) host
   cfg_file=/usr/local/nagios/etc/objects/localhost.cfg
   cfg_file=/usr/local/nagios/etc/objects/remoteClient.cfg
   ```

8. Open the `remoteClient.cfg` file and add the default host template name and define the remote hosts.

   ```
   vi /usr/local/nagios/etc/objects/remoteClient.cfg
   ```

   The following sample represents the file content:

   ```
   #HOST DEFINITION
   #Define a host for the local machine
   define host {
   use linux-server; Name of host template to use. This host definition will inherit all
   variables that are defined in (or inherited by) the linux-server host template
   definition.
       host_name   <host>
       alias       linux_client
       address     <ip address>
   }
   ################################################
   #
   # HOST GROUP DEFINITION
   # Define an optional hostgroup for Linux machines

   define hostgroup {

       hostgroup_name     linux-client  ; The name of the hostgroup
       alias              Linux Client  ; Long name of the group
       members            <host>  ; Comma separated list of hosts that belong to this group.
   }
   ```

   Replace the *host_name*, *alias*, and *address* with your remote host server details.

9. Add the services by removing the comments. When you remove the comments, these services are monitored in the Linux client machine where the Nagios Remote Plugin Executor is installed.

10. Add services based on the command present in the `nrpe.cfg` file in the client machine.

    `vi /usr/local/nagios/etc/nrpe.cfg` on the client machine where NRPE is installed.

    ```
    #Define a service to check the load on the local machine.
    define service{
        use                     local-service ;Name of service template to use
    ```

```
    host_name               <host>
    service_description     Current Load
    check_command           check_nrpe!check_load
        }
#Define a service to check the swap usage of the local machine.
define service {
   use                 local-service  ;Name of service template to use
   host_name               <host>
   service_description     Zombie Process Count
   check_command           check_nrpe!check_zombie_procs
    }
```

Note: To add more than one Linux host machine for monitoring, create a `remoteClient.cfg` configuration file for each machine and add it to the `nagios.cfg` file.

11. Create the NRPE command definition in the `commands.cfg` file.

    ```
    vi /usr/local/nagios/etc/objects/commands.cfg
    ```

12. Add the following NRPE command definition at the end of the file:

    ```
    #NRPE CHECK COMMAND
    #
    # Command to use NRPE to check remote host systems
    ################################################
    define command{
            command_name check_nrpe
            command_line $USER1$/check_nrpe -H $HOSTADDRESS$ -c $ARG1$
            }
    ```

13. Check that there are no errors in the configuration files.

    ```
    /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
    ```

14. Restart Nagios Core.

    ```
    service nagios restart
    ```

Step 4. Configure Nagios Core for Monitoring Dynamic Data Masking Installed on a Different Linux Machine

You can configure Nagios Core to monitor a Dynamic Data Masking server created on a different Linux machine.

1. Copy the sample code in "CheckServer.sh" on page 8 to create a `CheckServer.sh` file.

   Paste the content into a text editor and save to create the file.

2. Copy the `checkServer.sh` and `DDMDir.txt` files to the following location on the Dynamic Data Masking host machine:

   ```
   /usr/local/nagios/libexec/
   ```

3. In the `DDMDir.txt` file, add the paths to each Dynamic Data Masking installation. Do not add any other information. For example:

   ```
   /root/Informatica/<DDM991path>
   /root/Informatica/<DDM984path>
   /root/Informatica/<DDM251path>
   ```

4. Add the complete path to the DDMDir.txt file in the `checkServer.sh` file.

   ```
   #Provide full path to the file which contains DDM installed directories
   filename=/usr/local/nagios/libexec/DDMDir.txt
   declare -a DDM_PRESENT
   DDM_PRESENT=(`cat "$filename"`)
   ```

5. Run the following script:

   ```
   chmod +x /usr/local/nagios/etc/objects/checkServer.sh
   chmod +x /usr/local/nagios/etc/objects/DDMDir.txt
   ```

6. Add the custom plugin entries to the `nrpe.cfg` file.

```
### CUSTOM PLUGINS ###
command[check_server]=/usr/local/nagios/libexec/checkServer.sh
```

7. Restart the NRPE Service.

```
systemctl restart nrpe.service
```

8. On the Nagios Core Server, update the corresponding host configuration file. For example: `/usr/local/nagios/etc/objects/remoteClient.cfg`

```
#Define a service to check HTTP on the local machine.
# Disable notifications for this service by default, as not all users may have HTTP
enabled.

define service {

    use                     local-service           ; Name of service template to use
    host_name               <host name>
    service_description     DDM Processes
    check_command           check_nrpe!check_server
    contact_groups          admins
    contacts                nagiosadmin
}
```

9. Check the configuration file and ensure that there are no errors or warnings.

```
/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
```

10. Restart the Nagios Core server.

```
service nagios restart
```

Configure Nagios Core for Monitoring Dynamic Data Masking Installed on the Same Linux Machine

You can configure Nagios Core to monitor a Dynamic Data Masking server created on the same Linux machine where you install the Nagios Core server.

1. Copy the sample code in "CheckServer.sh" on page 8 to create a `CheckServer.sh` file.

   Paste the content into a text editor and save to create the file.

2. Copy the `checkServer.sh` and `DDMDir.txt` files to the following location on the Dynamic Data Masking host machine:

```
/usr/local/nagios/libexec/
```

3. Add the directory path where Dynamic Data Masking is installed in the DDMDir.txt file. Do not add any other information. For example:

```
/root/Informatica/<DDM991path>
/root/Informatica/<DDM984path>
/root/Informatica/<DDM251path>
```

4. Add the complete path to the DDMDir.txt file in the `checkServer.sh` file.

```
#Provide full path to the file which contains DDM installed directories
filename=/usr/local/nagios/libexec/DDMDir.txt
declare -a DDM_PRESENT
DDM_PRESENT=(`cat "$filename"`)
```

5. Run the following script:

```
chmod +x /usr/local/nagios/etc/objects/checkServer.sh
chmod +x /usr/local/nagios/etc/objects/DDMDir.txt
```

6. Edit the `/usr/local/nagios/etc/objects/commands.cfg` file to append the custom command.

```
define command {command_name check_server command_line /usr/local/nagios/etc/objects/
checkServer.sh }
```

7. Edit the `/usr/local/nagios/etc/objects/localhost.cfg` file to append the custom command.

```
# Define a service to check HTTP on the local machine.
# Disable notifications for this service by default, as not all users may have HTTP
enabled.

define service {

    use                     local-service           ; Name of service template to use
    host_name               localhost
    service_description     DDM Processes
    check_command           check_server
    contact_groups          admins
    contacts                nagiosadmin
                }
```

8. Check the configuration file and ensure that there are no errors or warnings.

```
/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
```

If the command generates permission issues, run the command `chmod -R 777 objects`. Path to the folder: `/usr/local/nagios/etc/objects`

9. Restart the Nagios Core server.

```
service nagios restart
```

CheckServer.sh

Use the following code to create a `CheckServer.sh` file:

```
#!/bin/bash

CURRENT=$(ps ax | grep -v grep | grep 'com.activebase.server.ServerConfigurationManager' | wc -
l)

DDM_RUNNING=( $(ps ax | grep -v grep | grep 'com.activebase.server.ServerConfigurationManager'
| awk '{print $(NF - 1)}') )

#Provide full path to the file which contains DDM installed directories
filename=<Enter the path to the DDMDir.txt file>
declare -a DDM_PRESENT
DDM_PRESENT=(`cat "$filename"`)

ddm_stopped=""
comma=", "
for ((i = 0 ; i < ${#DDM_PRESENT[@]} ; i++))
do
  flag=0
  for (( j = 0 ; j < ${#DDM_RUNNING[@]} ; j++))
  do
    if [[ "${DDM_RUNNING[$j]}" == *"${DDM_PRESENT[$i]}"* ]]; then
      flag=1
      break
    fi
  done
  if [[ $flag -eq 0 ]]; then
    ddm_stopped="$ddm_stopped${DDM_PRESENT[$i]}$comma"
  fi
done

if [[ "$CURRENT" == "${#DDM_PRESENT[@]}" ]]; then
  echo "OK - $CURRENT out of $CURRENT DDM are running"
  exit 0
elif [[ "$CURRENT" != "${#DDM_PRESENT[@]}" ]]; then
  echo "CRITICAL - DDM Not Running: $ddm_stopped"
  exit 2
else
  echo "WARNING - DDM Status Unknown"
  exit 1
fi
```

Configuring Nagios Core for Dynamic Data Masking on Windows

You can configure Nagios Core to monitor Dynamic Data Masking that you install on Windows machines.

The following sections describe the steps to configure Nagios Core to monitor Dynamic Data Masking on Windows machines. A single Nagios Core server can monitor multiple Dynamic Data Masking services.

Step 1. Install Nagios Core Server on a Linux Machine

To use Nagios Core for monitoring, you must install the Nagios Core server on a Linux machine.

You must have installed CentOS, Red Hat Enterprise Linux, or Fedora systems before you begin the steps. Log in as a root user to perform the install steps.

1. Run the following commands to install Apache and PHP and start the Apache service:

```
#yum install httpd php php-cli gcc unzip wget glibc glibc-common gd gd-devel net-snmp
#yum groupinstall "development tools"
#service httpd start
```

2. Run the following commands to create a Nagios Core user account and set the password:

```
#useradd nagios
#passwd nagios
#groupadd nagcmd
#usermod -a -G nagcmd nagios
#usermod -a -G nagcmd apache
```

   The code contains a sample user ID and password. Replace your User ID and password in the commands.

3. Run the following commands to download and install the latest Nagios Core service from the official web site:

```
# cd /opt/
# wget https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.4.5.tar.gz
# tar xzf nagios-4.4.5.tar.gz
# cd nagios-4.4.5
# ./configure --with-command-group=nagcmd
# make all
# make install
# make install-init
# make install-daemoninit
# make install-config
# make install-commandmode
# make install-exfoliation
```

4. Run the following command to configure Apache for Nagios Core:

```
#make install-webconf
#htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
#service httpd restart
```

5. Run the following commands to install Nagios plugins:

```
#cd /opt
#wget http://nagios-plugins.org/download/nagios-plugins-2.2.1.tar.gz
#tar xzf nagios-plugins-2.2.1.tar.gz
#cd nagios-plugins-2.2.1
#./configure --with-nagios-user=nagios --with-nagios-group=nagios
#make
#make install
#/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
#service nagios start
```

Step 2. Install NSClient++

NSClient++ is an agent that works with Nagios Core to monitor configured target systems.

SNMP must be enabled before you install the client. The following ports must be open on the firewall: 5666, 12489, and 5667. For more information on required ports, see the NSClient installation information.

1. Download the latest version of NSClient++ from the official website.
2. Run the installer.

   Choose **Generic** in step 1 and choose the **Typical** setup type.
3. In the **Allowed Hosts** field, define the IP address of the Nagios Core monitoring server.
4. Choose to load all modules. Under **Enable NRPE server**, choose **Insecure legacy mode**.
5. Complete the installation.
6. From the Services app, verify that the NSClient++ service is available.
7. Right-click the NSClient++ Service and open the **Properties** window.
8. Open the **Log On** tab.
9. Select the **Allow service to interact with desktop** option.
10. From the **Windows Features** window, verify that the SNMP is enabled and available.
11. From the **Services** app, right-click the SNMP Service and open the **Properties** window.
12. Open the **Log On** tab.
13. Select the **Allow service to interact with desktop** option.
14. Open the **Security** tab and add **Accepted community names**.

    Add the `public` community and assign Read and Create rights.
15. Restart the NSClient++ Service and the SNMP Service.

Step 3. Configure Nagios Core for Monitoring Dynamic Data Masking

Configure Nagios Core to monitor a Dynamic Data Masking server on Windows.

1. Open the following file: `C:\Program Files\NSClient++\nsclient.ini`
2. Uncomment all the modules defined in the Modules section.
3. Log in to the Nagios Core Server and add the object definitions to the Nagios configuration files to monitor the Windows machine.
   a. Open the following configuration file: `/usr/local/nagios/etc/objects/windows.cfg`

      ```
      vi /usr/local/nagios/etc/objects/windows.cfg
      ```

      The following sample is a Windows host definition. You can use the definition and replace the *host_name*, *alias*, and *address* values.

      ```
      # HOST DEFINITIONS
      #
      ###############################################
      # Define a host for the Windows machine that you want to monitor
      # Update the host_name, alias, and address values
      define host {
          use               <windows-server>  ;Inherit default values from a template.
          host_name         <host name>      ;The name that you assign to the host.
          alias             <add an alias>   ;A longer name associated with the host.
          address           <ip address>     ;IP address of the host
      ```

```
}
##############################################################
#
# HOST GROUP DEFINITIONS
#
##############################################################
# Define a hostgroup for Windows machines
# All hosts that use the windows-server template will automatically be a member of
this group
define hostgroup {
    hostgroup_name          windows-servers   ; The name of the host group
    alias                   Windows Servers   ; Long name of the group
}
```

b. Uncomment the following services that are already available in the `windows.cfg` file:

```
#Create a service for monitoring the version of NSClient++ that is installed.
#Change the host_name to match the name of the host you defined above.
#define service {
#    use                     generic-service
#    host_name               invww10dsg1317
#    service_description     NSClient++ Version
#    check_command           check_nt!CLIENTVERSION -s nagiosadmin
#}
# Create a service for monitoring the uptime of the server
# Change the host_name to match the name of the host you defined above
#define service {
#    use                     generic-service
#    host_name               invww10dsg1317
#    service_description     Uptime
#    check_command           check_nt!UPTIME -s nagiosadmin
#}
```

4. Open the `nagios.cfg` file.

   ```
   vi /usr/local/nagios/etc/nagios.cfg
   ```

5. Add an entry for the `windows.cfg` file to the main Nagios configuration file.

   ```
   #Definitions for monitoring a Windows machine
   cfg_file=/usr/local/nagios/etc/objects/windows.cfg
   ```

   To add multiple Windows host machines for monitoring, create a `windows.cfg` configuration file for each machine and add an entry for it to the `nagios.cfg`.

6. Review the `nagios.cfg` file for errors.

   ```
   /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
   ```

7. Restart Nagios Core

   ```
   service nagios restart
   ```

Set Up Email Notification Services on the Nagios Core Server

You can configure the Nagios Core server to send email notifications to required email IDs.

1. Browse to and open the following file: `/usr/local/nagios/etc/objects/contacts.cfg`
2. Add the following properties to the file:

```
define contact {
    contact_name         nagiosadmin             ; Short name of user
    use                  generic-contact         ; Inherit default values from generic-
contact template (defined above)
    alias                 Nagios Admin            ; Full name of user
    email                 abc@123.com     ; <<***** Add the required email address ******
    service_notification_period               24x7
    service_notification_options         w,u,c,r,f,s
    service_notification_commands        notify-service-by-email
    host_notification_period             24x7
```

```
    host_notification_options           d,u,r,f,s
    host_notification_commands          notify-host-by-email
}
```

The values for `service_notification_options` and `host_notification_options` represent different service and host states that you can configure for notifications. For example, you can choose to generate service notifications if the service enters a critical state, or host notifications when the host becomes unreachable. For information about service and host states that you can configure for notifications, see the Nagios Core documentation.

Update the *contact_name*, *alias*, and *email* fields with required values.

3. To add a group to which you want to send email notifications, add the following properties:

```
define contactgroup {
    contactgroup_name       admins
    alias                   Nagios Administrators
    members                 nagiosadmin
}
```

Update the *contactgroup_name*, *alias*, and *members* fields with required values.

4. Browse to and open the Linux configuration file in the following directory: `/usr/local/nagios/etc/objects/`
5. Add the contact group and contact name information:

```
define service {
    use                     generic-service
    host_name               invw10dsg1317
    service_description     DDM
    check_command           check_nt!PROCSTATE!-d SHOWALL -l ddm.exe -s nagiosadmin
    contact_groups          admins
    contacts                nagiosadmin
}
```

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